COMPILING STOCHASTIC CHEMICAL TRAJECTORIES WITH ELEMENTARY REACTIONS

HARJAS SANDHU

Abstract. We construct elementary mass-action networks that approximate the complete labelled trajectories of finite stochastic reaction lists of variable dimension and order. The input supplies a positive integer population weight and a global upper bound on its drift. The compiled structure and decoder depend only on the reaction list; all decoder columns are nonzero, and initialization uses exactly zero auxiliary molecules. Directed reachability is preserved for every fine state. One constant slowdown and rational coefficients give finite-horizon marked-path total-variation accuracy under arbitrary predictable coordinated multiplicative rate errors. Exact rescaled event times, original null labels and parallel labels are retained. A bounded pending-reaction credit and pathwise counting identities pay population and every fine firing throughout the actual untruncated run. Compiler bit costs and charged numerical physical costs have universal polynomial bounds, with reaction order included in the input. A separate paired construction supplies exact nominal detailed balance, conservation and the same timed-path guarantee for conservative reversible inputs with prescribed rational activities. The results concern ideal mass-action models; they do not construct molecular synthesis pathways or establish practical efficiency.

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1. Introduction

A stochastic chemical model can prescribe a reaction involving many molecules at once. An elementary implementation must instead assemble, transform and release those molecules through small steps. The intermediate reactions take time and create additional random events. Consequently, preserving a reaction's net stoichiometry does not establish that the implementation has the same observable dynamics.

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The question here is quantitative: can one replace a whole finite reaction list by reactions involving at most two molecules on each side, while approximating its complete timed, labelled trajectory and paying for all the intermediate work? We permit reaction order, dimension and initial populations to vary. The original system may be open, with no stationary distribution or conservation law. Its supplied population certificate is only a global upper drift bound.

1.1. **The contribution.** Theorem 2.1 gives such a compiler. Its species, channel structure and integer decoder depend only on the labelled stoichiometric list. Numerical rates, one slowdown and an error tolerance are then chosen from the full input. The initial auxiliary population is exactly zero. Every auxiliary molecule represents a nonempty multiset of original molecules, so there is no initially supplied controller or zero-payload fuel species.

Two conclusions are especially important. First, the graph statement holds from *every* fine state: directed reachability and communicating classes are exactly reflected by the decoder. Second, accuracy and actual work are uniform over predictable rate faults that may observe and use the entire fine history. The physical chain keeps running on rare excursions. Its population and all-event costs are bounded for every physical time, not just on the event where a finite-horizon comparison succeeds.

The core global estimate uses *pending molecules*. Each contains the true source payload of a reaction that may complete. Sparse binary collisions return pending molecules to their source representatives. A bounded credit assigns to each pending molecule an upper estimate of its chance to complete before returning. Its drift controls all completions, including null and parallel marks. Exact counting identities then pay even the rapid, unsuccessful assembly cycles. This mechanism is distinct from the finite-fibre averaging used for accuracy.

Theorem 9.1 gives a separate conservative, nominally detailed-balanced compiler. It uses paired source chains, different weights and different central rates. Conservation supplies its global work bound. The irreversible collision returns of the main construction are absent. Thus the balanced theorem is an additional construction, not a detailed-balance claim about the open compiler.

1.2. **Relation to previous methods.** Chemical programming and experimental molecular controllers are established subjects; programmable DNA controllers have been demonstrated by Chen et al. [2]. The present theorem is a statement about a specified stochastic mass-action model and its quantitative compilation contract.

Intermediate-species elimination and stochastic averaging are also established. Hoessly–Wiuf [3, Assumption 2 and Theorem 4.1] prove compact-time state-probability reduction under their non-interacting-species, fast-return and finite-closed-set hypotheses. Their Section 4.2 gives a time-dependent version with stated regularity. Here the target can have unbounded state space; the output retains the entire marked path, with finite rates, quantitative work bounds and history-dependent perturbations. We prove the required path estimate rather than infer it from convergence of marginals.

The factorial fibre law used below is standard product-form algebra. Anderson–Craciun–Kurtz [1, Theorem 4.1 and Section 4.2] describe Poisson invariant laws and the relation between chemical detailed balance and stochastic reversibility. We verify the finite-fibre identities directly. Spatial detailed balance requires further conditions: Zhang–Isaacson's bounded-domain analysis [4, Section III, Theorem III.1] treats reversible association–dissociation fluxes and product placement. Such spatial conditions are not consequences of a kinetic graph lift. Appendix C records the precise uses of these comparisons.

The contribution is the joint constructive guarantee: variable-order elementary lowering, all-state graph fidelity, canonical initialization, exact-time marked path approximation, predictable-fault robustness and a complete numerical and binary cost account. No claim of historical firstness is needed for that statement.

1.3. **Conceptual overview.** Figure 1 separates the actual operation from its analytical comparison. A reaction first assembles its source in a private chain. A pending species then either returns or completes into a private product chain. The physical system runs all these

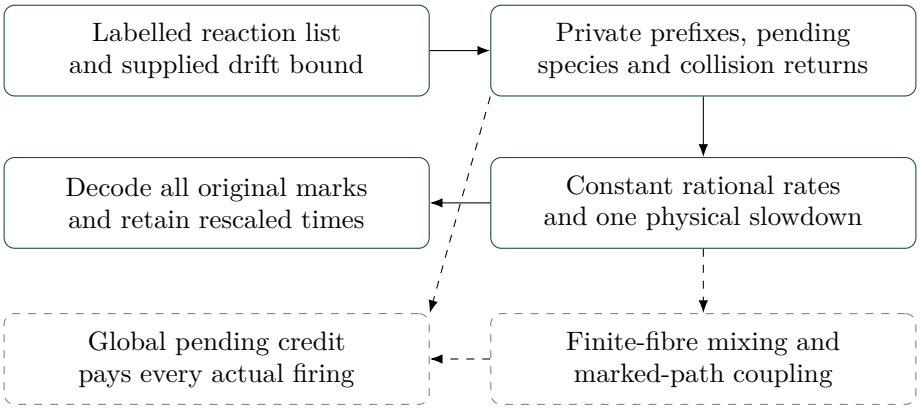


FIGURE 1. The upper four boxes describe the emitted operation. Dashed boxes are proof mechanisms. No cutoff, equilibrium sampler, proposal clock or externally performed reset is part of the physical network.

reactions autonomously with constant nominal coefficients. Observation applies a fixed decoder and hides only auxiliary labels.

For the proof, a weighted cutoff makes the relevant fibres finite. Hidden release estimates yield a uniform canonical atom, which gives a mixing integral. An independent proposal clock, used only in the coupling, compares full marked paths at their actual event times. The supplied drift bound controls the original process's exit probability. Independently, the pending credit controls actual work everywhere, including beyond that cutoff.

The reading order is as follows. Section 2 states the exact contract. Sections 3–4 construct the network and pay its whole run. Sections 5–6 prove timed-path accuracy. Section 7 and Appendix A establish the compiler and resource bounds. Section 8 gives examples, and Section 9 proves the balanced conservative interface.

2. Model, observation and main theorem

2.1. **Input and the drift promise.** Let $\mathbb{N} = \{0, 1, \ldots\}$. A finite labelled reaction list on d declared species $A_1, \ldots, A_d$ is

$$\mathcal{R} = ((y_r, z_r) : r = 1, \ldots, m), \qquad y_r, z_r \in \mathbb{N}^d.$$

Each complex is supplied as an explicitly expanded list of molecule indices; its order is therefore paid in the input length. Empty, null and parallel reactions are permitted. Empty sums and maxima are taken as zero. Put

$$q_r = |y_r|_1, \quad q'_r = |z_r|_1, \quad D = \max\bigl(1, \max_r (q_r \vee q'_r)\bigr).$$

For $x, y \in \mathbb{N}^d$, write

$$(x)_y = \prod_i x_i(x_i - 1) \cdots (x_i - y_i + 1),$$

with $(x)_0 = 1$ and $(x)_y = 0$ unless $x \geq y$. All propensities are at unit volume, with no additional symmetry factors. The original marked intensity is

$$\lambda_r(x) = k_r (x)_{y_r}, \qquad 0 < k_r \leq 1, \tag{1}$$

where the k_r are rational. A firing of label r changes x to $x - y_r + z_r$, even when this equals x.

The remaining input consists of binary initial counts x_0, positive integer weights ω_i, and rationals $A \geq 0$, $\tau > 0$, $0 < \epsilon < 1/8$. Set $N(x) = \omega \cdot x$ and $N_0 = N(x_0)$. The admission promise is

$$LN(x) = \sum_r k_r (x)_{y_r}\, \omega \cdot (z_r - y_r) \leq A \quad \text{for every } x \in \mathbb{N}^d. \tag{2}$$

The compiler is not required to decide this inequality. In particular, a certificate is supplied as a mathematical promise, not discovered by a normalizer, optimization or state-space oracle.

2.2. **Marked paths and physical time.** On reference time $[0,\tau]$, a complete original marked path is its initial state and the finite sequence

$$(t_j, r_j, x_j)_{j=1}^{J}, \qquad 0 < t_1 < \cdots < t_J \le \tau, \tag{3}$$

where x_j is the state just after the event. It retains a null event even when $x_j = x_{j-1}$, and distinguishes labels with identical increments. This record determines the piecewise constant state path and its final waiting interval. We use the usual Borel structure on the disjoint union of these finite-sequence spaces. For laws μ, ν, total variation means $d_{\mathrm{TV}}(\mu,\nu) = \sup_E |\mu(E) - \nu(E)|$.

A fine network has s species, elementary channels $\alpha_e \to \beta_e$ with $|\alpha_e|_1, |\beta_e|_1 \le 2$, an integer decoder $P \in \mathbb{N}^{d\times s}$, and a map $\ell(e) \in \{1,\dots,m\} \cup \{\dagger\}$. The symbol $\dagger$ denotes a hidden mark. Canonical initialization is $j(x) = (x, 0)$, and the original columns of P are the identity.

The emitted physical coefficients are constant rationals c_e. At physical time s, the actual channel intensity is

$$c_e a_e(s)(Z_{s-})_{\alpha_e}, \qquad 1 - \delta \le a_e(s) \le 1 + \delta. \tag{4}$$

The multipliers are arbitrary real predictable processes for a filtration in which the driving Poisson noise has independent future increments. They may coordinate using all prior fine states, hidden and observed marks, null and parallel events, and external randomness. No continuity or switching bound is assumed. They are faults, not chosen controls.

Given one slowdown $H \ge 1$, the observation $\mathrm{Obs}_{H,\ell}(Z)$ keeps PZ_0 and every fine firing with $\ell(e) \ne \dagger$, with physical event time divided by H, original label $\ell(e)$ and decoded post-event state. The observed physical interval is $[0, H\tau]$. We always use t for reference time and s for physical time.

Symbol	Meaning
x, X_t	Decoded/original state and target process in reference time.
u, Z_s	Fine state and actual process in physical time.
$n(u)$, $V(u)$, $N(Pu)$	Fine count $\lvert u\rvert_1$, unweighted decoded count $\lvert Pu\rvert_1$, and weighted decoded count.
P, j, ℓ	Fixed decoder, zero-auxiliary encoding and label map.
K, Q, η, H, δ	Chosen rational rate parameters, slowdown and fault tolerance.
B, Γ_B, W	Analytical cutoff, charged activity envelope and numerical charge.
$q(x)$, ν_x, I	Proof-only partition polynomial, fibre law and mixing integral.

The physical count, decoded count and weighted decoded count are not interchangeable. Chosen parameters and proof-only variables are distinguished in the guide above.

2.3. **Numerical and binary resources.** Let L be the complete binary length, including the species declarations, expanded complexes, counts, weights and all rational data. Define

$$B = 1 + \left\lceil \frac{32(N_0 + A\tau + 1)}{\epsilon} \right\rceil, \qquad U = 1 + B, \tag{5}$$

$$\Gamma_B = 2 + m + \sum_r k_r (1+B)^{q_r}, \tag{6}$$

$$W = 2 + d + m + D + \sum_i \omega_i + N_0 + A + \tau + \tau^{-1} + \epsilon^{-1} + \Gamma_B + \sum_r k_r^{-1}. \tag{7}$$

B is an analytical cutoff, not a reflecting wall. Γ_B is a computed rational activity envelope, not an oracle. Its binary length is polynomial in L, although its numerical magnitude may grow rapidly with order and population. The binary cost is charged in L; physical numerical costs are charged in W.

Theorem 2.1 (Open elementary trajectory compiler)**.** *There is a total deterministic algorithm with the following properties. For every labelled list $\mathcal{R}$, it constructs a fine species and channel structure, P and ℓ depending only on $\mathcal{R}$. Every decoder column is nonzero. Hidden channels preserve P, and every observed channel labelled r realizes an enabled original edge. For every pair of fine states,*

$$u \rightsquigarrow v \quad \Longleftrightarrow \quad Pu \rightsquigarrow Pv. \tag{8}$$

Its communicating classes are exactly $P^{-1}(\mathcal{C})$ as $\mathcal{C}$ ranges over the original communicating classes. For every numerical input above, the algorithm emits positive rational $c_e \le 1$, $H \ge 1$ and $0 < \delta \le 1/2$. If (2) holds, then under every *fault family (4), the fine process from $j(x_0)$ is nonexplosive for all physical times and*

$$d_{\mathrm{TV}}(\mathcal{L}(\mathrm{Obs}_{H,\ell}(Z)), \mathcal{L}(X \text{ marked on } [0,\tau])) \le \epsilon. \tag{9}$$

The same emitted output works for every such family. With $C = 2^{4096}$ and $p = 256$, binary work, memory and total output bits are at most $C(1+L)^p$, while

$$H,\ H\tau,\ \delta^{-1},\ \max_e c_e^{-1} \ \le CW^p. \tag{10}$$

Writing $F_{\mathrm{all}}(s)$ for all actual fine firings, including hidden and null firings, for every physical $s \ge 0$,

$$\mathbb{E} \sup_{0 \le v \le s} (1 + |Z_v|_1) \le CW^p(1+s), \tag{11}$$

$$\mathbb{E}F_{\mathrm{all}}(s) \le CW^p(1+s)^2, \tag{12}$$

$$\mathbb{E}F_{\mathrm{all}}(H\tau) \le CW^p. \tag{13}$$

The structure and graph assertions do not depend on the drift promise. The algorithm terminates on nonadmitted inputs as well.

The bounds are explicit sufficient bounds, not optimal exponents or measured performance. All labelled events and waiting times occur in the physical network. The conclusion approximates a path *law*; it is not an equality of laws or an exact-sampling claim.

2.4. **A preliminary consequence of the promise.**

Lemma 2.2 (Target escape and nonexplosion)**.** *The original marked process is nonexplosive and, for $b > N_0$,*

$$\mathbb{P}\left\{\sup_{t \le \tau} N(X_t) \ge b\right\} \le \frac{N_0 + A\tau}{b}. \tag{14}$$

Proof. Stop the minimal process at its first entrance into $N \ge b$. Below that level there are finitely many original states, and every marked rate is bounded. The finite reaction list also bounds the overshoot of N. Localized Dynkin's formula therefore gives $\mathbb{E}N(X_{t\wedge\sigma_b}) \le N_0 + At$. Nonnegativity yields (14). Letting $b \to \infty$ prevents population escape in finite time. An infinite number of firings with bounded population is impossible because the total intensity, including null labels, is bounded on each finite population set. This proves nonexplosion and removes localization. □

3. The elementary network

3.1. **Private prefixes and true payloads.**

Order the explicitly listed molecules of each complex deterministically. For label r, create source-prefix species $S_{r,j}$ for $2 \le j \le q_r$, product-prefix species $T_{r,j}$ for $2 \le j \le q'_r$, and a pending species C_r when $q_r > 0$. All these names are private, even when two prefixes have the same decoded multiset. In particular, a null reaction has separate source and product chains.

A prefix decodes to its first j original molecules; $PC_r = y_r$. Define the source representative s_r to be its original species if $q_r = 1$, or S_{r,q_r} otherwise. Define the product representative t_r

to be the empty complex if $q'_r = 0$, its original species if $q'_r = 1$, or T_{r,q'_r} otherwise. For any fine species S, put $d_S = |PS|_1$. Then

$$1 \le d_S \le D, \qquad n(u) := |u|_1 \le V(u) := |Pu|_1 \le N(Pu). \tag{15}$$

Thus an auxiliary is a representation of real decoded population, rather than a free token that disappears under observation.

For an ordered source $A_{i_1} + \cdots + A_{i_q}$, emit the hidden pairs

$$A_{i_1} + A_{i_2} \rightleftarrows S_{r,2}, \qquad S_{r,j-1} + A_{i_j} \rightleftarrows S_{r,j} \quad (3 \le j \le q), \tag{16}$$

and analogous hidden product pairs. For $q_r > 0$, add a hidden initiation and return pair, and an observed completion:

$$s_r \rightleftarrows C_r, \qquad C_r \xrightarrow{r} t_r. \tag{17}$$

For $q_r = 0$, the observed channel is $\varnothing \xrightarrow{r} t_r$. When $t_r = \varnothing$, this includes an actual $\varnothing \to \varnothing$ mark.

Finally, for every nonempty-source label r and *every* fine species S, emit the hidden collision return

$$C_r + S \longrightarrow s_r + S. \tag{18}$$

For $S = C_r$, its source is $2C_r$ and its propensity uses $(u_{C_r})_2$. The spectator is retained, including when it is another pending molecule. These reactions preserve the decoder and have at most two molecules on each side. Their small positive rate is essential for the global estimate.

Writing

$$S_{\mathrm{p}} = \sum_r (q_r - 1)_+ + \sum_r (q'_r - 1)_+, \qquad m_+ = |\{r : q_r > 0\}|,$$

the numbers of species and directed labelled channels are

$$s = d + S_{\mathrm{p}} + m_+, \qquad m_f = 2S_{\mathrm{p}} + 2m_+ + m + m_+ s. \tag{19}$$

The auxiliary populations are all initially zero. This construction uses only the labelled stoichiometry.

3.2. **Exact directed reachability.**

Proposition 3.1. *The network satisfies all graph assertions of Theorem 2.1.*

Proof. Hidden prefix steps and pending returns preserve the decoder, as does (18). If a nonempty-source completion can fire, its consumed C_r carries y_r, so $Pu \ge y_r$ and the decoded change is $z_r - y_r$. The assertion for an empty source is immediate.

From any fine state, first return each pending molecule through its baseline hidden return and then release every prefix. These unary releases terminate: they decrease the total number of bonds represented by prefixes. The result is $j(Pu)$. Every step used has a hidden reverse. Hence

$$u \longleftrightarrow j(Pu) \quad \text{in the hidden graph, for every } u. \tag{20}$$

An enabled original edge from x has a lift from $j(x)$: assemble its private source, enter C_r, fire its completion, and release the product. For an empty source, omit the first two operations. Consequently an original directed path lifts between canonical encodings. Together with (20), this proves the reverse implication in (8). Projecting a fine path and deleting hidden marks proves the forward implication. Every fine communicating class is therefore exactly $P^{-1}(\mathcal{C})$ for an original communicating class $\mathcal{C}$. The release paths in this proof are graph witnesses, not scheduled physical operations. □

Channel or hidden pair	Forward	Reverse
First source binding ($q_r \geq 2$)	$K\theta k_r U^{2-q_r}$	K
Later source binding	KU	K
First product binding	$K\theta U^{-2}$	K
Later product binding	K/U	K
$s_r \rightleftarrows C_r$ ($q_r \geq 2$)	K	K
$s_r \rightleftarrows C_r$ ($q_r = 1$)	$K\theta k_r$	K
Observed $C_r \to t_r$	Q	—
Observed $\varnothing \to t_r$	k_r	—
Collision return $C_r + S \to s_r + S$	ηd_S	—

TABLE 1. Raw coefficients for the open construction. A dash means that this channel does not carry a specified reverse. The first two rows apply to the private source chain and the next two to the private product chain.

3.3. **One explicit choice of rates.** For $m > 0$, abbreviate $\Gamma = \Gamma_B$ and $k_* = \min_r k_r$. Set

$$C_{\mathrm{f}} = 16(D+1)^2\big[1 + (\Gamma/k_*)^2\big], \qquad F = (D+1)\Gamma + 2Dm, \tag{21}$$

$$Q = 1 + 4096(1+m)U^3 C_{\mathrm{f}} + 128\epsilon^{-1}F\Gamma(1+\tau), \qquad \theta = Q^{-1}, \tag{22}$$

$$\Lambda = m + QU, \tag{23}$$

$$K = 1 + Q + 512\epsilon^{-1}BC_{\mathrm{f}}\Lambda(1+\Lambda\tau), \tag{24}$$

$$\eta = \frac{\epsilon}{128U^2(1+\tau)}, \qquad H = 1 + KU + Q + \eta D, \tag{25}$$

$$\delta = \frac{\epsilon}{128m_f U^2(1+H\tau)}. \tag{26}$$

All these quantities are rational. The raw coefficients κ_e are in Table 1; the emitted physical coefficients are

$$c_e = \kappa_e / H. \tag{27}$$

A common division of rates means that the nominal physical chain at time Ht has the raw chain's law at time t. No residence time is omitted.

Since $\theta m \leq 1$, every raw coefficient is at most $\max(KU, Q, \eta D, 1) < H$. Thus $0 < c_e \leq 1$, and $0 < \delta < 1/2$. The larger box $a_e \in [1/2, 3/2]$ will suffice for all global cost estimates. The smaller tolerance (26) is needed only for finite-horizon accuracy.

If $m = 0$, emit the original species only, $P = I$, no channels, $H = 1$ and $\delta = 1/2$. No minimum over rates or nonempty channel list is then evaluated. This case has the exact constant marked path.

4. The actual whole-run population and firing budget

This section concerns the physical, untruncated process under arbitrary predictable multipliers in $[1/2, 3/2]$. It does not use (2), a stationary law, or a successful-coupling event.

4.1. **Nonexplosion before the sharper estimates.**

Lemma 4.1. *The fine marked process is nonexplosive. On every finite physical interval its population has all finite polynomial moments, sufficient for the localized calculations below.*

Proof. A positive jump in $n = |Z|_1$ occurs only at a unary prefix release or a zero-source observed birth, and has size at most one. The total such intensity is at most $\frac{3}{2}(n+m)$: each prefix has one release of coefficient $K/H \leq 1$, and empty-source rates sum to at most m/H. A linear pure-birth process with immigration, initially $|x_0|_1$, dominates all positive population growth. It is nonexplosive and has finite polynomial moments on every finite interval. At population at most n, the total fine marked intensity is bounded by $\frac{3}{2}m_f(1+n)^2$. There can therefore be no accumulation of firings at bounded population, including hidden, collision and

null firings. This proves the assertions and justifies removing population localization after applying Dynkin's formula or compensator identities on a finite interval. □

4.2. **A bounded credit for pending reactions.** Write $\mathcal{L}^{\mathrm{raw}}_{\mathbf{a}}$ for the instantaneous generator before division by H, including the actual instantaneous multipliers. Define

$$G = \tfrac{3}{2}Q, \qquad \chi_r(v) = \frac{G}{G + K/2 + (\eta/2)(v - q_r)_+} \quad (q_r > 0), \tag{28}$$

$$\Psi(u) = \sum_{r:q_r>0} u_{C_r}\chi_r(V(u)), \qquad S_0 = 2D + \frac{4G}{\eta}. \tag{29}$$

A pending molecule has completion rate at most G. Its baseline and collision returns give a competing hazard that grows with decoded population. The credit χ_r is an upper completion probability for these competing hazards. It is not an additional species, a random decision rule, or a quantity computed during the physical run.

For $v \geq 2D$, $\chi_r(v) \leq 4G/(\eta v)$; for $v < 2D$, it is at most one. The number of pending molecules is at most V, so

$$0 \leq \Psi(u) \leq S_0, \qquad \Psi(j(x_0)) = 0. \tag{30}$$

Let

$$b(u, \mathbf{a}) = \sum_{r:q_r>0} Q a_r u_{C_r} \tag{31}$$

be the raw intensity of all nonempty-source completions.

Lemma 4.2 (Pointwise credit inequality)**.** *Put*

$$R_* = 1 + 4D + \frac{32GD}{\eta}, \qquad C_0 = 3KS_0 + GR_*^2. \tag{32}$$

For every fine state and every multiplier vector in $[1/2, 3/2]$*,*

$$\mathcal{L}^{\mathrm{raw}}_{\mathbf{a}}\Psi(u) \leq C_0 - \tfrac{1}{2}b(u, \mathbf{a}). \tag{33}$$

Proof. We account separately for removal, initiation and the changing credits of other pending molecules.

Removal and self-exclusion. The nominal sum of collision-return intensities targeting C_r is

$$\sum_S \eta d_S(u)_{e_{C_r}+e_S} = \eta u_{C_r}(V(u) - q_r). \tag{34}$$

Indeed the $S = C_r$ term is $\eta q_r u_{C_r}(u_{C_r} - 1)$, not $\eta q_r u_{C_r}^2$. Thus the actual per-molecule return hazard is at least $u_{min} = K/2 + (\eta/2)(V - q_r)$ whenever a pending molecule is present. If $g \leq G$ is its completion rate and $u \geq u_{min}$ its return rate, then

$$g(1 - \chi_r(V)) - u\chi_r(V) \leq 0.$$

Removing its credit therefore pays its completion intensity. No completion creates a pending molecule, because every product representative is original or belongs to a separate product chain.

Initiation. Each source representative of order at least two has one initiation with coefficient K. For a given original unary species, all its initiation coefficients sum to at most $K\theta\sum_r k_r \leq K$. Let $\chi_{max}(v) = \max_{q_r>0}\chi_r(v)$. The proof of (30) also gives $v\chi_{max}(v) \leq S_0$. The total initiation credit influx is at most $\frac{3}{2}KV\chi_{max}(V) \leq 3KS_0$. This estimate pays multiplicities of unary-source labels without an uncharged factor m.

Changes in surviving credits. An observed completion changes V by $q_r' - q_r$, of magnitude at most D. An increase of V only lowers the other credits; zero-source births have this favourable sign as well. For a decrease, first suppose $V \geq 4D$. Throughout the intervening interval $v \in [V - D, V]$,

$$|\chi_r'(v)| \leq \frac{8G}{\eta V^2}.$$

There are at most V surviving pending molecules. Their total credit gain per completion is consequently at most $8GD/(\eta V)$. If $V \geq R_*$ this is at most $1/4$, so the resulting positive drift is at most $b/4$. If $V < R_*$, each individual credit lies in $[0,1]$, so the gain is at most R_* per completion. Since $b \leq GV \leq GR_*$, its drift cost is at most GR_*^2. Other hidden transitions preserve V and either leave the pending counts unchanged or are the initiations and removals already paid. Adding the three estimates proves (33), with room to spare. □

The inequality is pointwise in the full rate box. Substituting a predictable random vector of multipliers therefore preserves it; no Markov assertion about the faulty marginal is involved.

Proposition 4.3 (Completions and population)**.** *Let $J_+(s)$ count nonempty-source observed completions and $J_0(s)$ all zero-source observed marks. For every physical $s \geq 0$,*

$$\mathbb{E}J_+(s) \leq \frac{2C_0}{H}s, \qquad \mathbb{E}J_0(s) \leq \frac{3m}{2H}s, \tag{35}$$

$$\mathbb{E}\sup_{v\leq s}(1+|Z_v|_1) \leq 1+|x_0|_1+\frac{D(2C_0+3m/2)}{H}s. \tag{36}$$

Proof. Divide (33) by H, apply Dynkin's formula, and use (30) and zero initial credit. Compensators give the first bound. Empty-source intensities sum to at most $3m/(2H)$, giving the second. Hidden channels preserve V, and each observed mark increases V by at most D. Hence pathwise

$$\sup_{v\leq s} V(Z_v) \leq |x_0|_1 + D(J_+(s)+J_0(s)).$$

Use (15) and take expectations. The preliminary nonexplosion and moment estimate in Lemma 4.1 justify the compensator and Dynkin calculations on every finite interval. □

4.3. **Exact counting identities.** Let A, R count actual prefix assemblies and releases, and let $\mathsf{I}, \mathsf{U}, \mathsf{C}, \mathsf{J}$ count pending initiations, baseline returns, collision returns and nonempty-source completions. Write F_0 for zero-source observed marks. All counts start at zero. Since no auxiliary molecule is supplied,

$$\sum_r Z_{C_r}(s) = \mathsf{I}-\mathsf{U}-\mathsf{C}-\mathsf{J} \geq 0. \tag{37}$$

Thus the four pending-related counts sum to at most $2\mathsf{I}$. An assembly lowers n by one and a release raises it by one. Nonempty-source completions cannot increase n, while an empty-source mark increases it by at most one. Nonnegativity of n implies

$$\mathsf{A} \leq |x_0|_1 + \mathsf{R} + \mathsf{F}_0. \tag{38}$$

These are pathwise identities or inequalities for the actual faulty run.

Proposition 4.4 (All-firing budget)**.** *For every physical $s \geq 0$,*

$$F_{\mathrm{all}}(s) \leq |x_0|_1 + 2\mathsf{R}(s) + 2\mathsf{I}(s) + 2\mathsf{F}_0(s), \tag{39}$$

$$\mathbb{E}F_{\mathrm{all}}(s) \leq |x_0|_1 + \frac{6K}{H}\int_0^s \mathbb{E}V(Z_v)\,dv + \frac{3m}{H}s. \tag{40}$$

Proof. Sum every fine channel count and apply (37) and (38). The total release intensity is at most $3Kn/(2H)$. The same is true of initiation, using $\theta m \leq 1$ for the unary groups and privacy for all other source representatives. The empty-source intensity is at most $3m/(2H)$. Compensators prove (40). □

Combining (36) and (40) gives linear expected population supremum and quadratic expected all-event growth for every physical time. In particular, the quadratic propensities in binding and collision channels are paid without a second-moment assumption on the target, or any stationary-law argument. Rare excursions have not been removed.

5. Finite fibres, factorial weights and mixing

For the accuracy proof, consider a nominal *reference* network obtained by omitting the collision returns only. This is an analytical comparison; the emitted network retains every collision-return channel with its positive rate. Temporarily suppress the reference network's observed channels as well.

5.1. **The exact hidden law.** Original species have relative activity one. The auxiliary relative activities are

$$w(S_{r,j}) = \theta k_r U^{j-q_r}, \qquad w(T_{r,j}) = \theta U^{-j}, \qquad w(C_r) = \theta k_r. \tag{41}$$

Write a fine state as (v, b), where b counts auxiliaries, E is their decoder matrix, and $v = x - Eb$ counts free original molecules in fibre $Pu = x$. For a vector z, $z!$ denotes the product of its coordinate factorials. Define

$$q(x) = \sum_{Eb \le x} \frac{w^b (x)_{Eb}}{b!}, \qquad \nu_x(v, b) = \frac{x!}{q(x)} \frac{w^b}{v!b!}. \tag{42}$$

Lemma 5.1 (Finite-fibre factorial identity)**.** *The hidden fibre is finite and irreducible, with unique invariant law ν_x. For $x \ge y_r$,*

$$\mathbb{E}_{\nu_x} C_r = \theta k_r (x)_{y_r} \frac{q(x - y_r)}{q(x)}. \tag{43}$$

The exact averaged observed intensity is

$$\bar{\lambda}_r(x) = k_r (x)_{y_r} \frac{q(x - y_r)}{q(x)}, \tag{44}$$

with value zero for disabled sources. For $y_r = 0$, the ratio is one.

Proof. Nonzero decoder columns bound $n(u)$ by $|x|_1$ on a fibre, so it is finite. The baseline hidden release paths in (20) prove irreducibility without using collision returns. The weights in (41) satisfy $\kappa_+ w^\alpha = \kappa_- w^\beta$ for each hidden pair. The first source binding has ratio $\theta k_r U^{2-q_r}$; each later binding multiplies the weight by U; the last pending exchange has ratio one. The product chain has successive ratios θU^{-2} and $1/U$. The unary pending pair has ratio θk_r.

For $u' = u - \alpha + \beta$, factorial cancellation gives

$$\frac{w^u}{u!} \kappa_+ (u)_\alpha = \frac{w^{u'}}{u'!} \kappa_- (u')_\beta.$$

The notation here includes weight one for originals. This calculation remains valid for a repeated-molecule first binding; no symmetry factor is missing. Restriction to the fibre and normalization gives (42).

To compute a pending mean, remove one C_r from the summation. Its weight is θk_r and its payload is y_r. The identity

$$(x)_{y_r + Eb} = (x)_{y_r} (x - y_r)_{Eb}$$

gives (43). Multiplication by $Q = \theta^{-1}$ gives (44). An empty-source mark has the constant raw rate k_r. Neither the partition sum nor an invariant distribution is evaluated by the compiler. □

5.2. **Partition bias below the analytical cutoff.** If $N(x) < B$, then $|x|_1 < B$ and

$$q(x) \le \exp\left(\sum_S w(S) B^{d_S} \right). \tag{45}$$

This follows by bounding $(x)_{Eb}$ by $B^{|Eb|_1}$ and dropping the summation constraint in (42). For a source prefix,

$$w(S_{r,j}) B^j \le \theta k_r U^{2j - q_r} \le \theta k_r U^{q_r}.$$

All source prefixes contribute at most $\theta D\Gamma$. Product prefixes contribute at most θDm, and pending species at most $\theta\Gamma$. Thus our slightly larger F in (21) gives

$$1 \le q(x) \le e^{\theta F}, \qquad N(x) < B. \tag{46}$$

The choice of Q gives $\theta F \le 1$. Both x and $x - y_r$ are below the cutoff when the source is enabled, so

$$|\bar{\lambda}_r(x) - \lambda_r(x)| \le 2\theta F \lambda_r(x). \tag{47}$$

The total original intensity there is at most Γ. A common-intensity coupling, retaining the original labels, gives stopped-path error at most

$$2\theta F \Gamma \tau \le \epsilon/64. \tag{48}$$

The precise stopping record is defined in Section 6.

5.3. **Release costs when the order varies.** A private prefix has only one upward continuation. One may tag an existing auxiliary copy analytically and follow its chain until it has completely released into original species. Distinct tags need not evolve independently. No physical particle label is introduced.

For a source of order $q \ge 2$, give $S_{r,2}, \dots, S_{r,q}, C_r$ depths $1, \dots, q$. Complete release is depth zero. Each downward step has rate K. Upward prefix rates are at most $KUB \le KU^2$; the last upward pending step has rate K. A reflecting birth–death chain on $\{0, \dots, q\}$, absorbed at zero, with upward rate Ku, $u = U^2$, and downward rate K dominates the release time. Its mean absorption time from the top is

$$\frac{1}{K} \sum_{j=0}^{q-1} (q-j) u^j \le \frac{q^2 U^{2q-2}}{K}. \tag{49}$$

For completeness, if T_i are the means and $d_i = T_i - T_{i-1}$, the equations at the reflecting top give $d_q = 1/K$, and the other equations give $d_i = 1/K + u d_{i+1}$. Summing these increments gives the displayed formula. The same comparison holds with time-dependent upward hazards bounded by Ku: extra upward arrows in a common-clock construction can only delay absorption.

Now $\Gamma \ge k_r U^q$, so $U^q \le \Gamma/k_*$. Equation (49) is at most C_{f}/K. For a product chain the upward hazard is at most $(K/U)B \le K$, giving mean release time at most D^2/K, even if no source has comparable order. A unary pending copy releases in mean time $1/K$. Thus, uniformly below the cutoff,

$$\mathbb{E}(\text{complete release time of an existing auxiliary lineage}) \le C_{\mathrm{f}}/K. \tag{50}$$

Paying source order through Γ and product order through the symmetric birth–death bound is essential. A large product order does not require an exponentially small physical coefficient $U^{-q'_r}$ on one channel.

Lemma 5.2 (Uniform hidden mixing)**.** *Let $F_x(t)$ be the nominal reference hidden kernel. With*

$$h = 16BC_{\mathrm{f}}/K, \tag{51}$$

for every $N(x) < B$ and every fine encoding u of x,

$$F_x(h; u, \{j(x)\}) \ge \tfrac{1}{2}, \tag{52}$$

$$d(t) := \sup_{N(x)<B,\, Pu=x} d_{\mathrm{TV}}(F_x(t; u, \cdot), \nu_x) \le 2^{-\lfloor t/h \rfloor}, \qquad I := \int_0^\infty d(t)\, dt \le 2h. \tag{53}$$

Proof. At most B auxiliary lineages are initially present. Equation (50), Markov's inequality and a union bound give probability at most $1/16$ that any survives to h. New lineages are born only by first source or product bindings and unary-source pending entries. Their total nominal rate is at most $4K\theta m U^2$ on the fibre. The probability of any such birth before h is at most

$$4K\theta m U^2 h = 64\theta m B U^2 C_{\mathrm{f}} \le 1/64,$$

by (22). If neither event occurs, the endpoint is $j(x)$. This proves (52). The argument did not assume independent service times or enumerate the fibre.

The common atom bounds the Dobrushin contraction coefficient of $F_x(h)$ by $1/2$. Iterating, and comparing a point mass with the invariant law, proves (53). Integration of the geometric block bound gives $h\sum_{j\ge 0}2^{-j}=2h$. □

6. Complete timed marked-path comparison

A path comparison must retain the interval before the first event, the hidden encoding after every event, and every original null mark. A state-only convergence theorem does not control these quantities. The following coupling provides the needed operation directly.

6.1. **A finite-speed averaging lemma.**

Lemma 6.1 (Marked uniformization)**.** *Consider a nominal fine Markov process with a decoder. Hidden transitions preserve its fibres and have kernels $F_x(t)$ with invariant laws ν_x. Assume*

$$\sup_{x,u:\,Pu=x} d_{\mathrm{TV}}(F_x(t;u,\cdot),\nu_x)\le d(t),\qquad I=\int_0^\infty d(t)\,dt<\infty.$$

Observed kernels $B_r(u,v)$ have total intensity at most Λ, and every r event sends the decoded state x to a specified $F_r(x)$. Kernels may include self-events and distinct labels with the same update. On a finite active state set, from any initial fine encoding of x_0, the complete marked output and the averaged chain of intensities $\bar b_r(x)=\sum_{u:Pu=x}\nu_x(u)\sum_v B_r(u,v)$ satisfy

$$d_{\mathrm{TV}}(\mathcal{L}(\textit{fine observed path}),\mathcal{L}(\textit{averaged path}))\le \Lambda I(1+\Lambda\tau). \tag{54}$$

The assertion also holds with absorbing exit records when the assumptions are required only on active fibres.

Proof. The case $\Lambda=0$ is immediate. Otherwise introduce a rate-Λ Poisson proposal clock, independently of the hidden dynamics, *in the coupling proof*. Between proposals run the hidden kernel. At a proposal from u, choose label r with probability $b_r(u)/\Lambda$, where $b_r(u)=\sum_v B_r(u,v)$, and otherwise reject. An accepted label uses the conditional landing kernel $B_r(u,\cdot)/b_r(u)$. This is an exact representation of the fine nominal process. Accepted null marks are real events; rejected proposals are not fine firings.

Condition on the complete coupled history at the previous proposal and the next gap of length a. While the decoded proposal histories agree at x, the fine endpoint has law $F_x(a;u,\cdot)$. Maximally couple that endpoint U' to a fresh variable V' of marginal ν_x. Their disagreement probability is at most $d(a)$. A conditional hidden bridge, if needed, completes the fine history over the interval without changing its marginal. Use common independent random marks for label acceptance and landing. If $U'=V'$, labels and their decoded landing states agree at the exact same proposal time.

Conditional on the previous joint history and the new gap, the comparison process accepts r with probability $\bar b_r(x)/\Lambda$. This probability is independent of the gap, so its marginal is exactly the averaged marked chain. After a proposal, including a rejection or null event, the actual fine encoding may be arbitrary; the next step still applies by uniformity of the mixing bound. After a coupling failure each marginal continues with its own transition law. On an exit, retain the exact label, time and landing state, then enter an absorbing record where all later proposals reject.

Write T_j for proposal times and $S_j=T_j-T_{j-1}$, with $T_0=0$. The probability of a coupling failure is at most $\mathbb{E}\sum_{j:T_j\le\tau}d(S_j)$. The first gap contributes $\int_0^\tau \Lambda e^{-\Lambda a}d(a)\,da$. For later gaps, the expected number of preceding proposal points before $\tau-a$ is $\Lambda(\tau-a)$, giving the renewal identity

$$\mathbb{E}\sum_{j:T_j\le\tau} d(S_j)=\int_0^\tau \Lambda e^{-\Lambda a}d(a)[1+\Lambda(\tau-a)]\,da\le \Lambda I(1+\Lambda\tau). \tag{55}$$

On success the entire observed records agree, not merely their endpoints. After the last proposal the hidden decoder remains constant; thus the final waiting layer needs no additional error. This proves the lemma. □

All proposal clocks, bridges, fresh invariant encodings and maximal couplings in this proof are analytical variables. The compiler implements none of them. In particular, no equilibrium sampler or tiny conditional probability is an input resource.

6.2. **Stopping records and nominal accuracy.** Stop a comparison at its first decoded entrance into $N \geq B$, retaining the exiting event's exact time, original label and decoded post-event state. This creates a finite active system with finitely many possible exit records: positive decoder weights bound pre-exit populations, and a finite list bounds exit increments. There is no assumption about mixing in an exit fibre. These stopped systems are used only in the proof.

The total raw reference central intensity before exit is at most $Q\sum_r C_r + m \leq QB + m < \Lambda$. Lemma 6.1 and (53) give

$$d_{\mathrm{TV}}(\text{reference stopped output}, \text{averaged stopped path}) \leq \frac{32BC_{\mathrm{f}}\Lambda(1+\Lambda\tau)}{K} \leq \epsilon/16. \tag{56}$$

The averaged/original error is (48). The common-intensity coupling for that estimate uses the actual label r and update $x - y_r + z_r$, so it also preserves each exiting time and landing state.

Now add back the emitted collision returns. Before exit, their raw nominal total intensity is

$$\eta \sum_{r:q_r>0} C_r(V - q_r) \leq \eta V^2 < \eta B^2.$$

A coupling of the reference and collision-augmented nominal fine processes until the first extra return has error at most

$$\eta B^2 \tau \leq \epsilon/128. \tag{57}$$

Thus omitting these reactions in the analytical reference did not omit their physical work or their finite-horizon effect.

6.3. **All-history fault comparison.** Compare the actual faulty and emitted nominal fine processes on physical $[0, H\tau]$, stopped at the decoded cutoff. Couple every fine mark, not just the observed ones. Until the first mismatch the full fine histories agree. At their common state the total unmatched intensity is at most

$$\sum_e c_e |a_e(s) - 1| (Z_{s-})_{\alpha_e} \leq \delta m_f U^2.$$

Indeed $n \leq V < B$ and every source order is at most two. Common marked Poisson thinning therefore gives

$$d_{\mathrm{TV}}(\text{faulty stopped fine path}, \text{nominal stopped fine path}) \leq \delta m_f U^2 H\tau \leq \epsilon/128. \tag{58}$$

The fault policy can use hidden and null marks and external randomness: while histories agree, its multipliers remain predictable in the coupled filtration. After a mismatch each marginal can continue with its own valid intensities. No Markov assumption on the faulty marginal is used.

Decoding, deletion of hidden marks and division of times by H are measurable maps and cannot increase total variation. Combining (48), (56), (57) and (58) yields

$$d_{\mathrm{TV}}(\text{faulty stopped decoded path}, \text{original stopped path}) \leq 3\epsilon/32. \tag{59}$$

6.4. **From stopped records to the full path.**

Lemma 6.2. *If two stopped marked-path laws, retaining the first exit event, have distance at most a, and the reference full path exits by the horizon with probability at most b, their full-path laws have distance at most $a + b$.*

Proof. Maximally couple the stopped records and extend each using its conditional full-path law. Such laws exist on the finite-sequence standard Borel path spaces. If the stopped records agree and the reference does not exit, both records are already their full paths, so they agree. Only stopped-record mismatch or reference exit can cause failure. The union bound proves the claim; it does not require a separate tail estimate for the other marginal. □

By Lemma 2.2 and (5), the original exit probability is less than $\epsilon/32$. Lemma 6.2 therefore gives full-path distance at most $\epsilon/8 < \epsilon$. This proves (9). The physical network has never stopped or reflected. Its actual excursions and all their costs are already paid by Propositions 4.3 and 4.4.

7. A terminating compiler and its resource ledger

The analytical proof used finite fibres, invariant distributions and couplings. The algorithm uses none of those objects. It performs the following finite operations.

(1) Read the declared species and labelled expanded complexes; order the molecules in each complex. Allocate the private prefixes and pending names. Emit their payload columns, the hidden pairs, each observed label, and all collision returns (18) over the allocated species list.

(2) If the label list is empty, emit the exceptional output specified in Section 3. Otherwise compute N_0, B, U, Γ_B and k_* by exact integer and rational arithmetic.

(3) Compute (21)–(26). Emit the coefficients in Table 1, divided by H, together with P, ℓ, H and δ.

It does not verify the admission promise or select a reaction trace. No population ball, communicating class or normalization constant is enumerated. The only variable powers are integer powers of explicitly bounded order.

Proposition 7.1 (Explicit numerical ledger)**.** *The parameters of the open compiler satisfy*

$$\begin{aligned} B, U &\le 2^7 W^3, & C_{\mathrm{f}} &\le 2^7 W^6, & F &\le 4W^2, \\ Q &\le 2^{42} W^{16}, & \Lambda &\le 2^{50} W^{19}, & K &\le 2^{125} W^{49}, \\ \eta^{-1} &\le 2^{22} W^8, & H &\le 2^{133} W^{52}, & m_f &\le 16 W^3. \end{aligned} \tag{60}$$

Moreover,

$$S_0 \le 2^{68} W^{24}, \qquad R_* \le 2^{71} W^{25}, \qquad C_0 \le 2^{196} W^{73}, \tag{61}$$

$$\max_e c_e^{-1} \le 2^{189} W^{74}, \qquad \delta^{-1} \le 2^{159} W^{63}, \qquad H\tau \le 2^{133} W^{53}. \tag{62}$$

For every physical time $s \ge 0$,

$$\begin{aligned} \mathbb{E} \sup_{v \le s} (1 + |Z_v|_1) &\le 2^{200} W^{74} (1+s), \\ \mathbb{E} F_{\mathrm{all}}(s) &\le 2^{210} W^{74} (1+s)^2, \\ \mathbb{E} F_{\mathrm{all}}(H\tau) &\le 2^{480} W^{180}. \end{aligned} \tag{63}$$

Proof. Appendix A gives the substitutions, including all small-rate reciprocals. For the whole-run estimates, insert (61) into (36). In (40) use $K/H \le 1$, and integrate the resulting linear population bound. Finally use $1 + H\tau \le 2^{134} W^{53}$ for the horizon estimate. No successful-path restriction enters these inequalities. □

Proposition 7.2 (Binary construction)**.** *The displayed compiler can be implemented using at most $2^{1000}(1+L)^{12}$ bit operations, working-memory bits and total output bits. The same bound suffices for the balanced construction in Section 9, using its complete input length.*

Proof. The declared-species and expanded-list format gives $d, m, D \le L$ and $S_{\mathrm{p}} \le L$. Hence $s \le 3L$ and $m_f \le 8(1+L)^2$. A dense output of stoichiometric vectors, decoder and labels is polynomial in L. The integer U has $O(L)$ bits. Every required power U^j, $j \le D$, has $O(L^2)$ bits. Addition of the rational terms of Γ_B does not multiply this by a state-space size: the product of all supplied rate denominators has total bit length at most L. The remaining formulas have fixed algebraic depth and degree. All numerators and denominators therefore have $O(L^2)$ bits.

More detailed conservative operation and bit-length allowances are given in Appendix A.4. They use schoolbook multiplication, division and Euclidean fraction reduction, and are dominated by the stated constant and power. There is no arithmetic operation on an exact unspecified real number. For the balanced variant the additional activity monomials and Ξ have integer exponents $O(D)$ and $O(L^2)$-bit rational representations as well. □

Propositions 3.1, 4.3, 4.4, 7.1, 7.2, and Section 6 complete the proof of Theorem 2.1. The deliberately larger common constants 2^{4096} and 256 dominate every displayed bound. A small binary output may still describe extremely slow or extremely active numerical dynamics. Those are different cost conventions.

8. Examples and necessary distinctions

8.1. **A genuinely open nonlinear input.** Consider the one-species list

$$\varnothing \xrightarrow{\alpha} A, \quad A \xrightarrow{\beta} \varnothing, \quad 2A \xrightarrow{\gamma} 3A, \quad 3A \xrightarrow{\rho} 2A, \qquad 0 < \alpha, \beta, \gamma, \rho \le 1. \tag{64}$$

All four rates are rational. There is no positive conserved linear population weight. For $N(n) = n$,

$$LN(n) = \alpha - \beta n + \gamma n(n-1) - \rho n(n-1)(n-2) \le \alpha + (\gamma + 3\rho)n^2 - \rho n^3.$$

The maximum of $cn^2 - \rho n^3$ over real $n \ge 0$ is $4c^3/(27\rho^2)$. Thus a valid supplied certificate is

$$A = \alpha + \frac{4(\gamma + 3\rho)^3}{27\rho^2}. \tag{65}$$

For instance $\alpha = \beta = \rho = 1$, $\gamma = 1/2$ meets all requirements and has no common detailed-balanced activity: the birth–death pair would require activity one, whereas the other pair would require activity $1/2$. The theorem therefore applies beyond conservative or balanced inputs.

For the $2A \to 3A$ label, the emitted private chains include

$$2A \rightleftarrows S_{r,2} \rightleftarrows C_r \xrightarrow{r} T_{r,3}, \qquad 2A \rightleftarrows T_{r,2}, \quad T_{r,2} + A \rightleftarrows T_{r,3}.$$

The collision channels $C_r + S \to S_{r,2} + S$ are also present for every fine species S. Decoding C_r gives $2A$, while decoding $T_{r,3}$ gives $3A$. No three-molecule fine reaction has been emitted. The observed transition is correct whenever it fires, while its finite-rate path law is approximate.

8.2. **Null and parallel labels carry information.** Take $x_0 = 3A$ and two distinct original labels

$$3A \xrightarrow{r_1} 3A, \qquad 3A \xrightarrow{r_2} 3A$$

with rates $k_1, k_2 > 0$. The state path is constant. Nevertheless the labelled counts are independent Poisson variables with means $6k_1\tau$ and $6k_2\tau$. Deleting the second label creates marked-path total-variation error at least $1 - e^{-6k_2\tau}$, although the state-path error remains zero. Pooling the labels can likewise preserve total event rate while losing the marked record. This explains why null and parallel labels are present in the path space, the coupling, the fault filtration and every firing budget.

In the compiler, a null completion can change the fine encoding even though it leaves the decoded state unchanged. Its post-event mixing layer must therefore be paid just like that of a non-null event.

8.3. **Why the physical clock cannot be free.**

Proposition 8.1 (A null-label clock lower bound)**.** *Fix $D \ge 3$ and the single original label $DA \to DA$ of rate one. Consider any elementary implementation with m_f channels, coefficients at most one, nonzero nonnegative integer decoder columns and the required hidden and observed*

decoder changes. If its marked path from $x_0 = NA$ on reference $[0,1]$ has total-variation error at most $1/8$, then, for sufficiently large N,

$$H \geq \frac{3N^{D-2}}{2^{D+5} m_f}. \tag{66}$$

Proof. At zero faults, the original count is Poisson of mean $\lambda = (N)_D$. For $\lambda \geq 32$, Chebyshev's inequality gives $\mathbb{P}\{J \geq \lambda/2\} \geq 7/8$. The approximation therefore has at least $3/4$ probability of that many observed events. Its expected all-firing count is at least $3\lambda/8$.

Every fine firing preserves the decoded population N, so nonzero decoder columns give fine count at most N throughout the run. Each elementary channel has intensity at most $(N+1)^2$. Thus its expected total count through physical H is at most $m_f(N+1)^2 H$. For $N \geq 2D$, $\lambda \geq (N/2)^D$ and $(N+1)^2 \leq 4N^2$. Combining the inequalities gives (66). □

The structure, hence m_f, is fixed before N is chosen. A population-independent clock guarantee would therefore be false. This lower bound does not contradict the charged Γ_B and polynomial slowdown of Theorem 2.1, nor does it assert optimality of its constants.

9. A separate conservative detailed-balanced compiler

This section gives the kinetic interface for conservative spatial realizations. The reaction list now has a supplied reverse-label involution $r \mapsto \bar{r}$, with $y_{\bar{r}} = z_r$ and $z_{\bar{r}} = y_r$. A fixed label must be null. Supplied positive rational activities a_i satisfy

$$k_r a^{y_r} = k_{\bar{r}} a^{z_r}. \tag{67}$$

The positive integer weight is *conserved*:

$$\omega \cdot (z_r - y_r) = 0 \quad \text{for every } r. \tag{68}$$

Use (5)–(7) with $A = 0$, and define the additional activity charge

$$\Xi = 2 + \left(2 + \sum_i (a_i + a_i^{-1})\right)^{2D+2}, \qquad W_{\mathrm{b}} = W\big|_{A=0} + \Xi. \tag{69}$$

The complete binary length now includes the activities and reverse-label list. Ξ explicitly pays large and small activity scales; its numerical value is not assumed polynomial in D or in the logarithms of the a_i.

Theorem 9.1 (Balanced conservative interface)**.** *For the inputs (67)–(68), a total deterministic compiler emits an elementary network, P, ℓ and a reverse-channel involution depending only on $\mathcal{R}$ and $r \mapsto \bar{r}$. All decoder columns are nonzero, the original columns are identity, canonical initialization is exactly $(x_0, 0)$, and the all-state directed reachability and class assertions of Theorem 2.1 hold.*

It emits rational $0 < c_e \leq 1$, $H \geq 1$, $0 < \delta < 1/2$ and positive rational fine activities $\widetilde{a}_S$, with $\widetilde{a}_i = a_i$ on original species, such that every actual reverse pair satisfies

$$c_e \widetilde{a}^{\alpha_e} = c_{\bar{e}} \widetilde{a}^{\beta_e}. \tag{70}$$

The reverse of a hidden mark is hidden, and the reverse of a mark r is $\bar{r}$. Every admitted predictable fault family has full marked-path accuracy (9); detailed balance is asserted nominally, at zero faults.

For every physical time and fault history,

$$|Z_s|_1 \leq \omega \cdot P Z_s = N_0. \tag{71}$$

Writing m_f for the emitted channel count, the process is nonexplosive and

$$\mathbb{E} F_{\mathrm{all}}(s) \leq \tfrac{3}{2} m_f (1 + N_0)^2 s \quad (s \geq 0). \tag{72}$$

With universal $C_{\mathrm{b}} = 2^{1024}$, $p_{\mathrm{b}} = 128$, binary work, memory and output are at most $C_{\mathrm{b}}(1+L)^{p_{\mathrm{b}}}$, and

$$H + H\tau + \delta^{-1} + \max_e c_e^{-1} + \max_S (\widetilde{a}_S + \widetilde{a}_S^{-1}) + \max_{i,S} P_{iS} \leq C_{\mathrm{b}} W_{\mathrm{b}}^{p_{\mathrm{b}}}. \tag{73}$$

The population supremum, all-time firing and horizon bounds (11)–(13) *hold with these constants and* W_{b} *as well.*

If $m = 0$, emit the original species, no channels, $P = I$, the original activities, $H = 1$ and $\delta = 1/4$. All conclusions are immediate, with empty channel maxima taken as zero. The rest of this section assumes $m > 0$.

9.1. **The paired elementary network.** For each directed label with $q_r > 0$, create only a private source chain $S_{r,2}, \dots, S_{r,q_r}$ and a pending species C_r. Their decoder columns are their true source prefixes and y_r. There is no separate product chain: the reverse label provides it. Keep the hidden prefix pairs and $s_r \rightleftarrows C_r$ as before, but omit every collision-return channel. For label r, the observed channel is

$$\begin{cases} C_r \to C_{\bar{r}}, & q_r, q_{\bar{r}} > 0, \\ \varnothing \to C_{\bar{r}}, & q_r = 0 < q_{\bar{r}}, \\ C_r \to \varnothing, & q_{\bar{r}} = 0 < q_r, \\ \varnothing \to \varnothing, & q_r = q_{\bar{r}} = 0. \end{cases} \tag{74}$$

The middle two cases display the general paired formula. In the conservative input class they cannot occur, since a positive conserved weight precludes $0 \leftrightarrows z \neq 0$. A self-reverse nonempty null label is the actual channel $C_r \to C_r$. Distinct labels remain distinct.

Put $J = \sum_r q_r$. There are exactly

$$s = d + J, \qquad m_f = 2J + m. \tag{75}$$

Each hidden pair contributes two directed channels; each original label contributes one observed channel. The reverse involution is therefore explicit, including self-reverse null channels. Release to $j(Pu)$ and canonical lifting prove the all-state graph statement exactly as in Proposition 3.1: a central event now lands in the reverse-side pending species, which is then released. Every step conserves $\omega \cdot P$, so (71) holds under arbitrary rate errors, and (72) proves nonexplosion and pays every actual firing.

9.2. **Weights, coefficients and the nominal balance certificate.** Use the following relative weights, different from (41):

$$w(S_{r,j}) = \theta U^{-j}, \qquad w(C_r) = \theta U^{-q_r}. \tag{76}$$

Set $\widetilde{a}_S = w(S)a^{PS}$ for auxiliaries and $\widetilde{a}_i = a_i$ for originals. Raw rates are

Channel	Raw coefficient
First prefix binding	$K\theta/U^2$
Later prefix binding	K/U
Every prefix release	K
$s_r \to C_r$, $q_r \geq 2$	K
$A_i \to C_r$, $q_r = 1$	$K\theta/U$
Every baseline $C_r \to s_r$	K
Observed r, $q_r > 0$	$g_r = Qk_rU^{q_r}$
Observed r, $q_r = 0$	k_r

The first and later hidden binding ratios give precisely (76); multiplication by the payload monomial a^{PS} therefore proves the activity equality for each hidden pair. For a nonempty central source,

$$g_r\widetilde{a}_{C_r} = Qk_rU^{q_r}\theta U^{-q_r}a^{y_r} = k_r a^{y_r} = k_{\bar{r}}a^{z_r}. \tag{77}$$

This proves central balance against its actual reverse. The same calculation handles an empty side in (74). For empty null pairs, (67) gives $k_r = k_{\bar{r}}$; fixed null channels satisfy balance identically. Common division by H proves (70) exactly, not approximately.

Choose

$$\begin{aligned} C_{\mathrm{f,b}} &= 4(D+1)^2, \\ Q_{\mathrm{b}} &= 1 + 4096(1+m)UC_{\mathrm{f,b}} + 128\epsilon^{-1}(J+1)\Gamma(1+\tau), \\ \theta_{\mathrm{b}} &= Q_{\mathrm{b}}^{-1}, \qquad \Lambda_{\mathrm{b}} = m + Q_{\mathrm{b}}\Gamma U, \\ K_{\mathrm{b}} &= 1 + Q_{\mathrm{b}}\Gamma + 512\epsilon^{-1}BC_{\mathrm{f,b}}\Lambda_{\mathrm{b}}(1+\Lambda_{\mathrm{b}}\tau), \\ H_{\mathrm{b}} &= 1 + K_{\mathrm{b}} + Q_{\mathrm{b}}\Gamma, \qquad \delta_{\mathrm{b}} = \frac{\epsilon}{128 m_f U^2 (1+H_{\mathrm{b}}\tau)}. \end{aligned} \tag{78}$$

In the preceding rate table use these balanced parameters, and emit $c_e = \kappa_e / H_{\mathrm{b}}$. Below, subscripts are suppressed when discussing this construction alone. Since $k_r U^{q_r} \le \Gamma$, every raw coefficient is at most $\max(K, Q\Gamma, 1) < H$. Thus $0 < c_e \le 1$.

9.3. **Fibre law and timed-path accuracy.** The factorial proof of Lemma 5.1, with (76), gives the same partition expression and

$$\mathbb{E}_{\nu_x} C_r = \theta U^{-q_r} (x)_{y_r} \frac{q(x - y_r)}{q(x)}. \tag{79}$$

The factor a^x cancels on conditioning on $Pu = x$. Multiplication by $g_r = Qk_r U^{q_r}$ again gives the exact averaged rate (44). This change in central coefficient is necessary; the open construction's Q alone would give the wrong average here.

There are J auxiliaries, each contributing at most θ to the exponential bound, since $w(S)B^{|PS|_1} \le \theta$. Consequently $1 \le q(x) \le e^{\theta J}$ below the cutoff, and the averaged/original marked-path error is at most $2\theta J\Gamma\tau \le \epsilon/64$.

For hidden mixing, every tagged prefix has downward hazard K and upward hazard at most $(K/U)B \le K$; the full-prefix/pending exchange adds at most one depth. The recurrence used in (49), now with upward/downward ratio one, bounds its mean release time by $(D+1)^2/K \le C_{\mathrm{f,b}}/K$. At most B auxiliary lineages are present initially. New first bindings and unary pending entries have total rate at most $K\theta m$. At $h = 16BC_{\mathrm{f,b}}/K$, the probability of an initial survivor is at most $1/16$ and that of a new lineage at most $16\theta m B C_{\mathrm{f,b}} < 1/64$. Thus there is a common canonical atom of mass at least $1/2$ and $I \le 2h$, with no state enumeration.

The central raw intensity is at most $m + Q\Gamma B < \Lambda_{\mathrm{b}}$. Lemma 6.1 therefore gives error at most $32BC_{\mathrm{f,b}}\Lambda_{\mathrm{b}}(1+\Lambda_{\mathrm{b}}\tau)/K_{\mathrm{b}} \le \epsilon/16$. It permits the arbitrary reverse-pending encoding after an observed event and the unchanged decoded state after a null event. The all-fine-mark fault coupling from (58) costs at most $\epsilon/128$. Conservation prevents exit in either process, since $N_0 < B$. The full marked-path error is consequently at most $11\epsilon/128 < \epsilon$. This proves accuracy without importing any unproved general-open balanced tail estimate.

9.4. **Uniform activities, costs and initialization.** Let $M_a = 2 + \sum_i (a_i + a_i^{-1})$. Every payload has order at most D, so a^{PS} and its reciprocal are at most $M_a^D \le \Xi$. Also $U^{q_r} \le \Gamma/k_*$. It follows directly that

$$\max_S (\widetilde{a}_S + \widetilde{a}_S^{-1}) \le 2Q\Gamma k_*^{-1}\Xi. \tag{80}$$

These activities certify the rates; they are not the actual initial counts. Initialization remains exactly $j(x_0)$, with no auxiliary molecules. Appendix A.3 gives all numerical substitutions and binary precision costs. In particular,

$$\begin{aligned} Q &\le 2^{25} W_{\mathrm{b}}^6, & \Lambda &\le 2^{33} W_{\mathrm{b}}^{10}, \\ K &\le 2^{88} W_{\mathrm{b}}^{27}, & H &\le 2^{89} W_{\mathrm{b}}^{27}, \\ \delta^{-1} &\le 2^{113} W_{\mathrm{b}}^{37}, & \max_e c_e^{-1} &\le 2^{128} W_{\mathrm{b}}^{39}, \\ \max_S (\widetilde{a}_S + \widetilde{a}_S^{-1}) &\le 2^{26} W_{\mathrm{b}}^9. \end{aligned}$$

Equations (71)–(72) give the claimed all-time bounds, and $\mathbb{E}F_{\mathrm{all}}(H\tau) \le 2^{94}W_{\mathrm{b}}^{32}$. The binary argument is Proposition 7.2. The universal balanced constants therefore cover the entire interface, proving Theorem 9.1.

Corollary 9.2 (The zero-upper-drift balanced subcase)**.** *If a reversible input has the supplied positive rational activity certificate* (67) *and the promise $LN \le 0$ on every integer state, then the supplied positive integer ω is conserved by each reaction. Hence Theorem 9.1 applies without an additional conservation promise.*

The proof is given in Appendix B.1. It uses the finite complex graph, not enumeration of the original state space.

9.5. **A small balanced example.** For $2A \rightleftarrows B$ with both rates one, take $a_A = a_B = 1$ and $\omega = (1, 2)$. The private auxiliaries are $S_{+,2}, C_+, C_-$, with payloads $2A, 2A, B$. The observed pair is $C_+ \rightleftarrows C_-$ with raw coefficients QU^2, QU, respectively. Its activity products are $(QU^2)(\theta/U^2) = 1 = (QU)(\theta/U)$. All hidden pairs satisfy the same exact check. The conserved decoded population is $x_A + 2x_B$, even though a prefix can represent two original molecules in one fine particle.

10. Scope and further questions

The output is an autonomous stochastic mass-action model with specified species, stoichiometry and constant nominal coefficients. It is a mathematical compilation theorem. It does not discover actual molecular species that realize these reactions, synthesize them, control off-graph leakage, or calibrate experimental rates. A laboratory realization would have additional constraints on preparation, separation of species, transport, energy, waste, stability and measurement that are not premises of this theorem.

The universal binary cost and the universal numerical physical cost answer different questions. The latter pays Γ_B, small input rates and the requested accuracy and horizon. It can be large even when every binary representation is short. The constants are sufficient mathematical bounds and do not establish an economical implementation, an optimal exponent, or a matched practical advantage over direct simulation or existing chemical controllers.

The trajectory statement is also specific. It retains exact rescaled event times in an approximation of full path laws. It does not assert equality of the two laws. It is stronger than a state-marginal or jump-chain comparison, but does not yield an exact equilibrium sampler. Initial and post-event waiting layers are included; no trace is selected or postselected.

Several precise questions remain. Can the same broad open input class be compiled with nominal detailed balance and polynomial all-time costs under arbitrary predictable faults? The conservative balanced result does not settle that question; reversing collision returns changes the credit calculation (Appendix B). Can sharper resource bounds approach the clock lower bound of Proposition 8.1? Can useful molecular implementations realize a substantial part of this contract with realistic leakage and preparation costs? These questions require additional operations, not a relabelling of the present compiler.

The balanced conservative theorem supplies an explicit kinetic input for spatial work: elementary paired channels, rational activities, a nonzero payload decoder, canonical zero-auxiliary initialization and quantitative marked-path and work guarantees. Reflecting diffusion, finite-radius contact rules and spatial placement balance remain separate mathematical constructions. No general open spatial realization follows from Theorem 2.1.

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The mathematical ideas, constructions, proofs, and manuscript were developed by GPT-6 Astra, with additional AI assistance for review. Human input was minimal and consisted primarily of setting the research objectives and providing occasional feedback.

Appendix A. Explicit numerical and binary estimates

This appendix spells out the substitutions behind the universal resource claims. In every inequality $W \geq 2$, and each nonnegative summand in (7) is at most W.

A.1. **Open parameters and the whole-run ledger.** From the ceiling in (5),

$$B \leq 2 + 32\epsilon^{-1}(N_0 + A\tau + 1) \leq 98W^3, \qquad U \leq 99W^3 \leq 2^7W^3.$$

Also $(D+1)^2 \leq 4W^2$ and $1 + (\Gamma/k_*)^2 \leq 2W^4$. Hence $C_{\mathrm{f}} \leq 2^7W^6$, while $F \leq 4W^2$. The two nonconstant terms of Q are bounded respectively by $2^{41}W^{16}$ and $2^{10}W^5$. Thus $Q \leq 2^{42}W^{16}$ and $\Lambda = m + QU \leq 2^{50}W^{19}$. Since $1 + \Lambda\tau \leq 2^{51}W^{20}$, the last term of K is at most $2^{124}W^{49}$; adding $1 + Q$ gives $K \leq 2^{125}W^{49}$. Next

$$\eta^{-1} = 128U^2(1+\tau)\epsilon^{-1} \leq 2^{22}W^8.$$

The definition of H gives $H \leq 2^{133}W^{52}$. From $S_{\mathrm{p}} \leq 2mD$, (19) gives $m_f \leq 16W^3$. This proves (60).

Since $G \leq 2^{43}W^{16}$, the dominant terms of S_0 and R_* are bounded by $2^{67}W^{24}$ and $2^{70}W^{25}$. Thus

$$S_0 \leq 2^{68}W^{24}, \quad R_* \leq 2^{71}W^{25}, \quad C_0 \leq 3\,2^{193}W^{73} + 2^{185}W^{66} \leq 2^{196}W^{73}.$$

Use $|x_0|_1 \leq N_0 \leq W$ and $H \geq 1$ in (36) to obtain $\mathbb{E}\sup_{v\leq s}(1 + n(Z_v)) \leq 2^{200}W^{74}(1+s)$. In (40), $K/H \leq 1$, so integration gives

$$\mathbb{E}F_{\mathrm{all}}(s) \leq W + 6\,2^{200}W^{74}(s + s^2/2) + 3Ws \leq 2^{210}W^{74}(1+s)^2.$$

Finally $1 + H\tau \leq 2^{134}W^{53}$ gives $\mathbb{E}F_{\mathrm{all}}(H\tau) \leq 2^{478}W^{180} \leq 2^{480}W^{180}$. These are estimates for the actual untruncated process.

A.2. **Every small raw coefficient is paid.** For a source first binding with order $q \geq 2$,

$$(K\theta k_r U^{2-q})^{-1} \leq QU^{q-2}/k_r \leq Q\Gamma k_*^{-2} \leq 2^{42}W^{19}.$$

For a product first binding its reciprocal is at most $QU^2 \leq 2^{56}W^{22}$. Unary pending initiation has reciprocal at most $Q/k_* \leq 2^{42}W^{17}$; a later product binding at most U; a collision return at most η^{-1}, since $d_S \geq 1$. Other hidden and observed reciprocals are at most 1 or k_*^{-1}. Therefore

$$\max_e \kappa_e^{-1} \leq 2^{56}W^{22}, \qquad \max_e c_e^{-1} \leq H\,2^{56}W^{22} \leq 2^{189}W^{74}.$$

Likewise,

$$\delta^{-1} = 128m_fU^2(1+H\tau)\epsilon^{-1} \leq 2^{159}W^{63}.$$

The product-prefix weights θU^{-j} may be much smaller than any individual physical coefficient. Only adjacent ratios are emitted; charging a whole long product weight as a required reaction rate would give an unnecessary order-dependent cost.

A.3. **Balanced parameter ledger.** Use W_{b} throughout this subsection. The cutoff bounds are unchanged, $J \leq mD \leq W_{\mathrm{b}}^2$, $m_f \leq 4W_{\mathrm{b}}^2$, and $C_{\mathrm{f,b}} \leq 2^4W_{\mathrm{b}}^2$. The two nonconstant Q_{b} terms are at most $2^{24}W_{\mathrm{b}}^6$ and $2^9W_{\mathrm{b}}^5$, giving $Q_{\mathrm{b}} \leq 2^{25}W_{\mathrm{b}}^6$. Hence $\Lambda_{\mathrm{b}} \leq 2^{33}W_{\mathrm{b}}^{10}$ and $1 + \Lambda_{\mathrm{b}}\tau \leq 2^{34}W_{\mathrm{b}}^{11}$. Substitution in K_{b} yields a last term at most $2^{87}W_{\mathrm{b}}^{27}$, and therefore

$$K_{\mathrm{b}} \leq 2^{88}W_{\mathrm{b}}^{27}, \quad H_{\mathrm{b}} \leq 2^{89}W_{\mathrm{b}}^{27}, \quad H_{\mathrm{b}}\tau \leq 2^{89}W_{\mathrm{b}}^{28}.$$

It follows that

$$\delta_{\mathrm{b}}^{-1} \leq 128(4W_{\mathrm{b}}^2)(2^{14}W_{\mathrm{b}}^6)(2^{90}W_{\mathrm{b}}^{28})W_{\mathrm{b}} = 2^{113}W_{\mathrm{b}}^{37}.$$

The reciprocal of a balanced first-binding raw coefficient is at most $QU^2 \leq 2^{39}W_{\mathrm{b}}^{12}$. Other reciprocals are bounded by QU, U, 1 or k_*^{-1}, and so $\max_e c_e^{-1} \leq 2^{128}W_{\mathrm{b}}^{39}$. Equation (80) gives the activity sum bound $2^{26}W_{\mathrm{b}}^9$; decoder entries are at most $D \leq W_{\mathrm{b}}$.

For all actual faults, fine population is at most N_0. The deterministic upper bound on marked intensity is

$$\tfrac{3}{2}m_f(1+N_0)^2 \leq 2^5W_{\mathrm{b}}^4.$$

Consequently $\mathbb{E}F_{\mathrm{all}}(H\tau) \le 2^{94}W_{\mathrm{b}}^{32}$, and all-time population and firing estimates have respectively constant and linear time factors. The sum in (73) and all these costs are covered by $2^{1024}W_{\mathrm{b}}^{128}$.

A.4. **Binary lengths and a concrete arithmetic implementation.** Use binary integers and reduced numerator–denominator pairs. The complete input length pays every declared species, label and listed molecule, so $d, m, D \le L$ and the total number of source and product molecule entries is at most L. Sorting each complex by its integer species index and accumulating its dense prefix payloads requires polynomial work; no isomorphism or network normal form is computed.

The integer N_0 is a sum of products of input integers, whose total input bit length is at most L. The ceiling calculation of B uses a fixed number of rational products and one integer division. In particular, U has at most an absolute constant times $1 + L$ bits. All powers U^j, $0 \le j \le D$, can be computed successively and have at most an absolute constant times $(1 + L)^2$ bits. In Γ_B, a common denominator dividing the product of the input rate denominators suffices. Its bit length is at most L; thus the sum still has $O((1 + L)^2)$-bit numerator and denominator.

The expressions for $C_{\mathrm{f}}, F, Q, \Lambda, K, \eta, H, \delta$ have a fixed number of rational operations and fixed degree in their intermediate variables. Taking maxima or minima of finitely many input rates requires only rational comparisons. The output first-binding powers are among the already computed integer powers of U or their reciprocals. Even if every fraction is reduced after each operation, all intermediate integer lengths are bounded by $2^{20}(1 + L)^2$ bits. One can check this allowance using the rational height $h(p/q) = \max(\log_2(1 + |p|), \log_2(1 + q))$: addition, multiplication and division have height at most the sum of operand heights plus one, and an integer power j has height at most $jh + 1$. The input sums and the integer U have height at most $32(1 + L)$; the common-denominator argument above gives $h(\Gamma_B) \le 64(1 + L)^2$. Expanding the displayed fixed parameter formulas uses fewer than 2^{12} such height contributions, including the remaining powers of U. Their sum fits $2^{20}(1 + L)^2$.

There are at most $3L$ fine species and $8(1 + L)^2$ open channels. Writing dense source and product vectors, the dense decoder and all output rates therefore requires at most $2^{20}(1 + L)^4$ arithmetic or copying operations on those integers. The paired construction has at most $d + \sum_r q_r$ species and $2\sum_r q_r + m$ channels, and satisfies the same allowances. For its activities, compute a^{PS} from the listed payload, or from prefix products. The sum of input activity denominator lengths is at most L, and every exponent is at most D. Both these monomials and the $O(D)$-th power defining Ξ have $O((1 + L)^2)$-bit representations. They fit the preceding integer allowance.

On b-bit integers, schoolbook multiplication and long division use $O(b^2)$ bit operations. Euclidean reduction, even using the crude bound of $O(b)$ divisions, uses $O(b^3)$. With the allowances just given, the entire operation count is bounded by a constant less than 2^{1000} times $(1 + L)^{10}$, hence by $2^{1000}(1 + L)^{12}$. Working memory and total output are smaller than the same bound. This establishes Proposition 7.2 with explicit slack; it is an implementation bound, not a report that a full compiler or stochastic physical realization was executed.

Appendix B. Balance, conservation and stationary-output boundaries

B.1. **Why zero upper drift forces conservation in the balanced class.**

Proof of Corollary 9.2. Let Y_i be independent Poisson variables of means a_i. This is an analytical test law; no equilibrium initialization is performed. The factorial moment identity gives $\mathbb{E}(Y)_{y_r} = a^{y_r}$. Pairing reverse labels with (67) therefore gives

$$\mathbb{E}LN(Y) = \sum_r k_r a^{y_r} \omega \cdot (z_r - y_r) = 0.$$

The polynomial LN is integrable under this law. Since $LN(x) \le 0$ for every integer state and every state has strictly positive Poisson mass, $LN(x) = 0$ for every $x \in \mathbb{N}^d$.

A polynomial vanishing on $\mathbb{N}^d$ is identically zero, by induction in the coordinates and the one-variable root bound. The multivariate falling-factorial polynomials are linearly independent: in one coordinate they are monic of distinct degrees, and their products form the corresponding triangular multivariate basis. Grouping coefficients by source complex c yields

$$\sum_{r:y_r=c} k_r\,[\omega\cdot z_r-\omega\cdot c]=0.$$

Thus $c\mapsto\omega\cdot c$ is harmonic on the finite reverse-connected complex graph, with positive edge weights k_r. At a maximum in a connected component, every nonpositive difference in the last sum must vanish. Connectivity propagates this equality throughout the component. Hence $\omega\cdot z_r=\omega\cdot y_r$ for every label, proving conservation. □

B.2. **The partition correction is real.** For the balanced fine network, the product-Poisson law with means $\widetilde{a}_S$ has exact pairwise stochastic balance by the factorial calculation in Lemma 5.1. All positive-weight conservation classes are finite; their normalized restrictions are stationary, and their mixture is stationary as well. This is the familiar product-form situation of [1], with its hypotheses explicitly fulfilled here.

The decoded weights of that product law, before an overall constant, are

$$\frac{a^x}{x!}q(x). \tag{81}$$

Indeed summation over $v+Eb=x$ factors out a^x and leaves exactly the partition sum (42). Its correction cannot in general be erased. For an unconditioned fine product-Poisson law, the decoder generating function is

$$\mathbb{E}z^{PZ}=\exp\left(\sum_S \widetilde{a}_S(z^{PS}-1)\right). \tag{82}$$

Original-species activities are fixed at a_i. Any positive auxiliary of payload order at least two adds a positive higher monomial to the logarithm, which cannot cancel. Unit-payload auxiliaries add positive linear terms. Therefore an unconditional decoded law with positive auxiliaries is not exactly the original product-Poisson law with the same prescribed original activities. This observation concerns the unconditioned law; it is not a blanket claim about every conditioned conservation class. Finite-horizon trajectory approximation only needs the small, explicitly paid partition bias.

B.3. **Why a reversible graph does not inherit the open credit.** The open compiler's return $C_r+S\to s_r+S$ has coefficient ηd_S. Detailed balance would require a reverse coefficient $\eta d_S\widetilde{a}_{C_r}/\widetilde{a}_{s_r}$. That reverse creates pending molecules with quadratic population dependence. The initiation estimate in Lemma 4.2 pays a linear influx; its proof does not apply to this altered process. Conservation in Theorem 9.1 resolves the whole-run issue for its own input class. For general open balanced inputs, reversing the collisions alone is not the missing global argument.

Nor does a common frozen equilibrium justify arbitrary feedback. On states $0,1$, let a frozen parameter $b\in[1-\delta,1+\delta]$ multiply both unit rates. Each such generator is reversible for the uniform law. Choose $b=1+\delta$ in left-limit state 0 and $b=1-\delta$ in state 1. The resulting predictable rule gives rates $0\to1$ equal to $1+\delta$ and $1\to0$ equal to $1-\delta$, with stationary mass $(1+\delta)/2$ at 1. This does not obstruct all reversible compilers; it identifies an invalid invariance shortcut. The fault proof in Section 6 instead couples all fine marks and uses a uniform intensity bound.

Appendix C. Exact roles of the primary comparisons

No central estimate is imported as an unspecified averaging or mixing service. The graph lift, finite-fibre law, canonical atom, marked uniformization, predictable-fault comparison and whole-run budgets are proved in this paper. The primary comparisons have the following limited roles.

Intermediate-species reduction. Hoessly–Wiuf [3, Assumption 2, Theorem 4.1] use a finite closed set, non-interacting species, proper fast reactions, their propensity assumption and a fast-return identity, with initial law on states free of the eliminated species. Their conclusion is uniform compact-time $O(\varepsilon)$ state-probability reduction. Section 4.2 imposes smooth time-dependent rates. Neither the open input class nor the entire marked-law and global-work contract here is asserted as an application of that theorem.

Product-form algebra. In [1, Theorem 4.1], a positive complex-balanced deterministic equilibrium gives product-Poisson stationary weights and normalized restrictions to closed irreducible classes. Section 4.2, equations (4.8)–(4.9) and Theorem 4.5 discuss detailed balance and stochastic reversibility. Our hidden fibres are finite, connected and have an explicit positive pairwise activity certificate, so factorial cancellation proves the required identity directly. No target equilibrium or quantitative mixing rate is taken from that result.

Spatial flux balance. Zhang–Isaacson [4, Section III, equation (III.2) and Theorem III.1] consider one reversible $A + B \leftrightarrows C$ association–dissociation system with stated reaction kernels, free configuration domain and reflecting boundary conditions. Pointwise forward and reverse spatial flux balance gives their equilibrium formulas; product-placement rules near boundaries require additional attention in Section IV. We use this as context for separating kinetic activity balance from spatial realization, not as an all-network adaptive compiler theorem.

Experimental chemical programming. The DNA controllers of Chen et al. [2] are an experimental precedent for chemical programming. Our formal elementary species and chosen rational coefficients do not establish the molecular implementation, calibration, leakage behaviour or experimental performance of those controllers.

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The University of Queensland, Brisbane, Australia
Email address: harjas.sandhu@student.uq.edu.au